

Video Game Programming: Game Analytics

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Motivations

- Since thousands of games are published yearly, it is difficult to attract players' attention
- Social functions make it very easy for players to promote a game or destroy its reputation
- Understanding players' behaviors in the game is thus fundamental to improve players' experience
- In addition to questionnaires and physiological measurements, game analytics is an emerging resource to collect data on players' behavior and analyze it to extract useful patterns

Game data collection

- The first step to perform game analytics is to collect game data
- Data concerning players' profiles and in-game events are typically sent to remote databases
- A typical flow is sending structured data (e.g., in json) to a dynamic Web page (e.g., in PHP) that performs SQL queries on a database (e.g., in MySQL) using custom servers or cloud services
- It is important to pay attention to players' privacy (e.g., consent, anonymization) and possible network issues (e.g., no connection, network errors), which require a local temporary backup

Pre-processing

- After collecting data and before applying techniques for the analysis, the collected data must be pre-processed as suggested in Srivastava et al., Web Usage Mining: Discovery and Applications of Usage Patterns from Web Data, ACM SIGKDD Explor. Newsl., 1, 2, 12-23, 2000
- The CRISP process (see Drachen et al., Game Data Mining, in Game Analytics, 205-253, 2013) also remarks that game data should be cleaned from errors (e.g., due to network errors) and transformed to a technique-friendly format
- E.g., PFI game data cleaning is described in Chittaro and Buttussi, Learning Safety through Public Serious Games: A Study of "Prepare for Impact" on a Very Large, International Sample of Players, TVCG, 28, 1573-1584, 2022

Data mining

- Data mining is the extraction of implicit and interesting patterns from large data collections (see Klossgen and Zytkow, Handbook of data mining and knowledge discovery, 2002)
- Reviews in educational context (e.g., Romero and Ventura, Educational Data Mining: A Review of the State of the Art, IEEE Trans Syst Man Cybern Part C, 40, 6, 601-618, 2010) include up to a dozen of groups of techniques, some of which can be applied to games
- Techniques include statistical methods, AI algorithms (e.g., of machine learning) for supervised and unsupervised pattern extraction, and data visualization

Statistics

- Statistics range from reporting metrics (e.g., mean, standard deviation) to applying tests to compare different populations or measures assessed at different times (e.g., pre-post)
- In HCI, statistical tests are typically used to analyze measures acquired during studies in the laboratory
- Statistical tests can be applied also to big data sets acquired during studies in-the-large, including game data
- In this case, it is fundamental to consider the **effect sizes** and not only the statistical significance, because even small differences can be statistically significant in big data sets

Classification

- Classification techniques assign instances (e.g., players) to classes
- They are supervised methods, because the analysts should decide the classes before the algorithm calculates the assignments considering a set of variables
- The C4.5 algorithm can generate a decision tree to predict the class for each instance and the tree will represent the rules extracted from data
- Useful tools to apply C4.5 are **Weka** and **Keel**
- Another classification technique is Self-Organizing Map, which is based on neural networks

Clustering

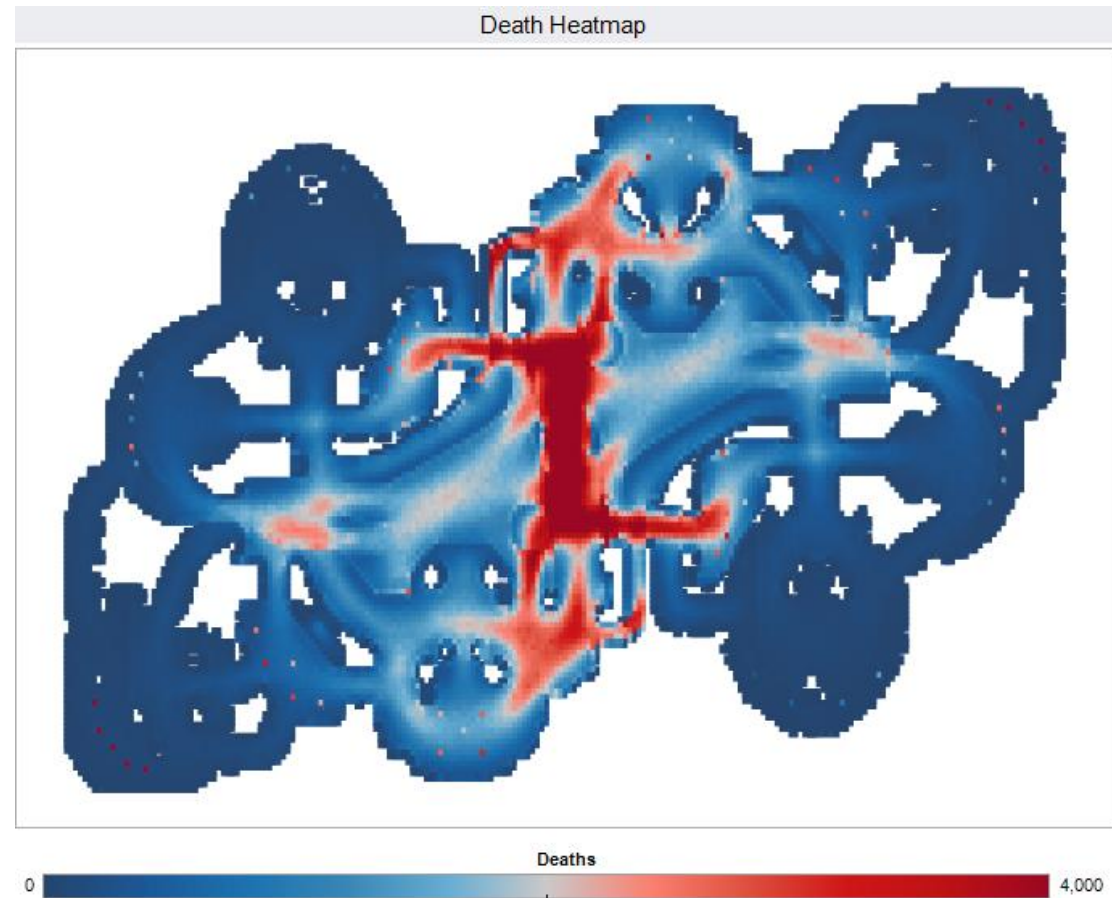
- Clustering techniques organize instances in clusters without having information on final clusters
- They are unsupervised: most of them (e.g., **k-means**) ask the analyst only the number of clusters and determine properties automatically
- The properties of a cluster can be expressed as the values of a representative instance (e.g., the **centroid**)
- Tools as Weka include a k-means implementation used in e-learning
- K-means is sometimes unsuited to analyze game data, because centroids of different clusters may look similar
- Other clustering techniques (e.g., **archetypal analysis**) are more suited because they use "extreme" instances as representative of the clusters

Association rules and recurring patterns

- Association rules look for associations between variables expressed as if-then rules (e.g., if players have score higher than X in level Y, then they have score higher than Z in level W)
- Weka includes an implementation of the Apriori algorithm to find such rules
- A technique variant focuses on recurring patterns, e.g., errors that are usually made together (see Merceron and Yacef, Mining Student Data Captured from a Web-Based Tutoring Tool: Initial Exploration and Results, J. Interact. Learn. Res., 15, 4, 319-346, 2004)
- APrioriAll can be used to find recurring patterns considering also the temporal sequence (see Agrawal e Srikant, Mining sequential patterns, Procs 11th Intl Conf Data Engineering, 3-14, 1995)

Visualization

- Data visualization allows to display (game) data in a way that makes it easier for the analysts to detect interesting patterns
- Visualizations range from simple graphs to complex (interactive) visualizations (e.g., using maps as in fig.)



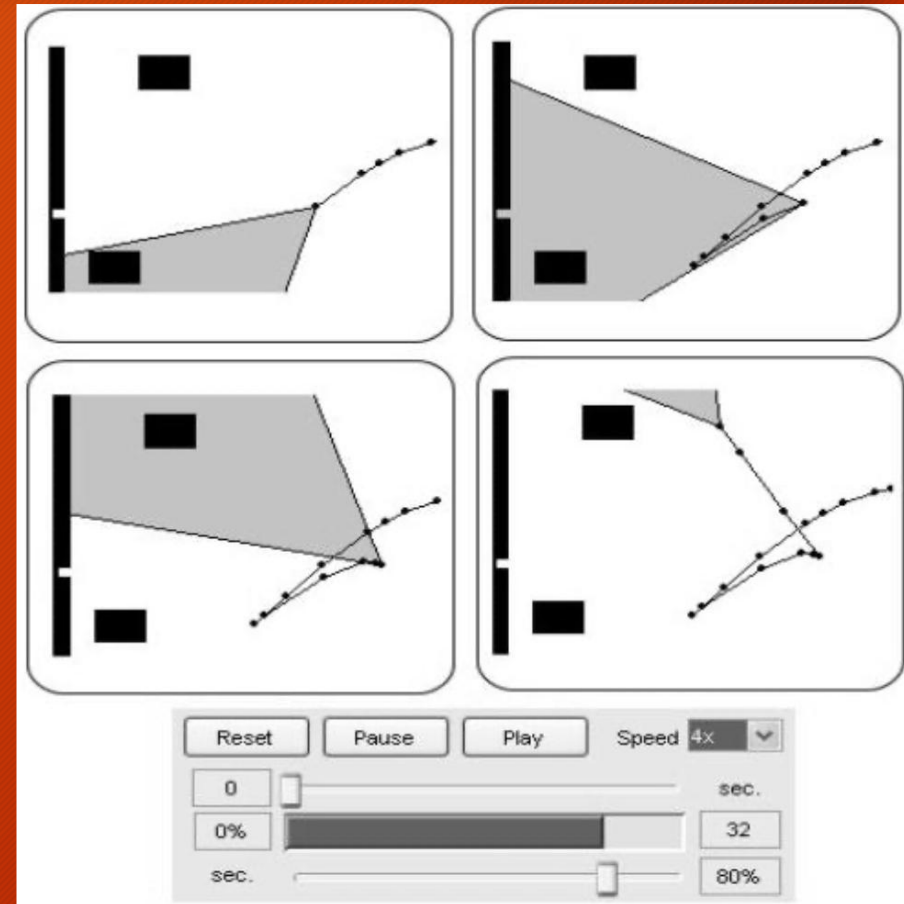
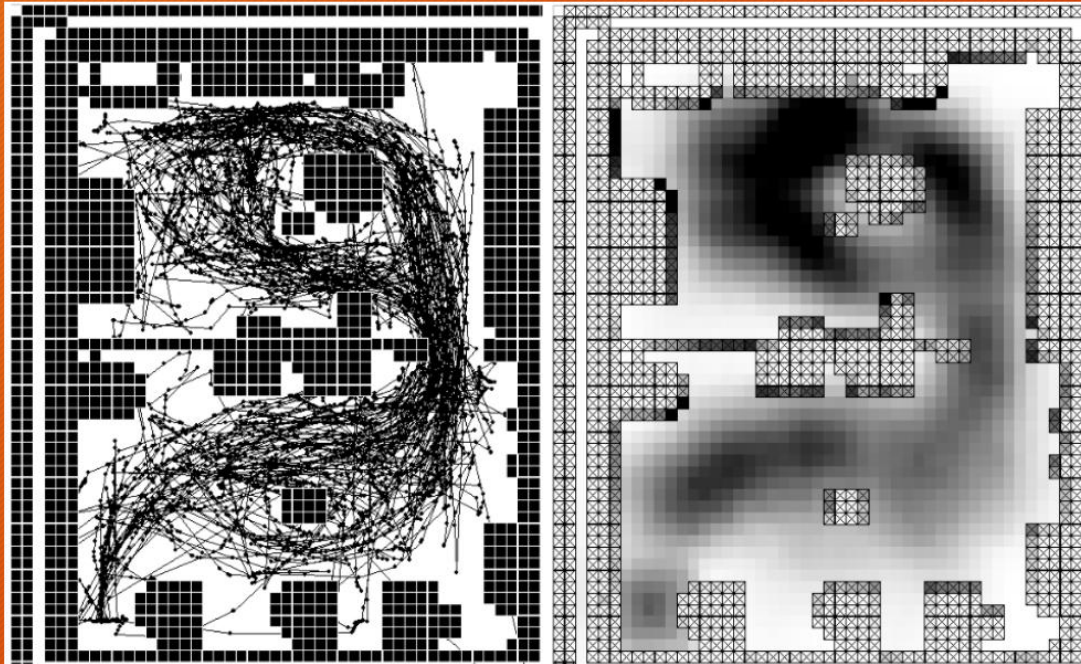
Death heatmap of Transformers: War for Cybertron from <https://gameanalytics.com/blog/balance-and-flow-maps/>

VU-Flow 1/6

- Tool proposed by Chittaro & Ieronutti, 2004; Ieronutti et al., 2005
- Detection:
 - Interests of users (most viewed objects)
 - Navigation problems
 - Users' visiting style
- Data gathered: position and orientation
- Visualizations:
 - Time-dependent visualization
 - Areas where users stayed more/less time
 - Time-independent visualization
 - Users' paths
 - Areas more/less traveled
 - Areas more/less seen
 - Users' predominant flow

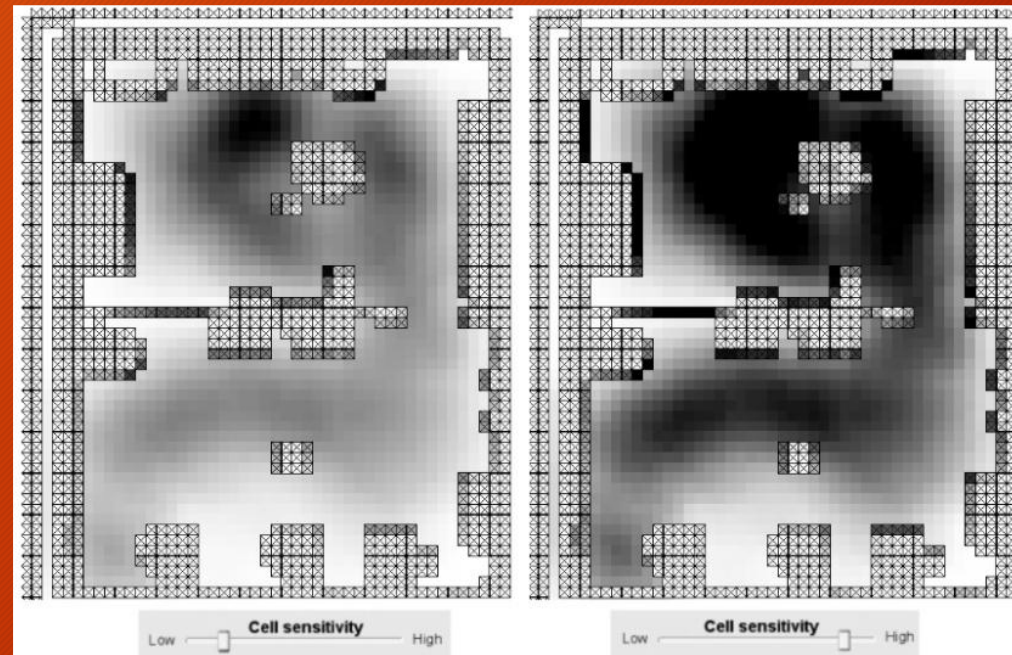
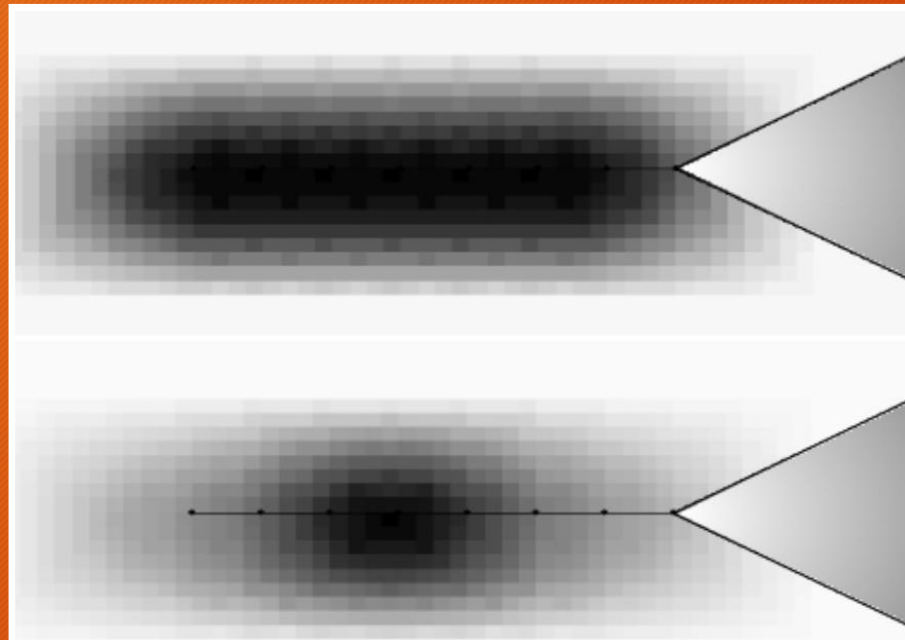
VU-Flow 2/6

- Travelled areas & users' paths



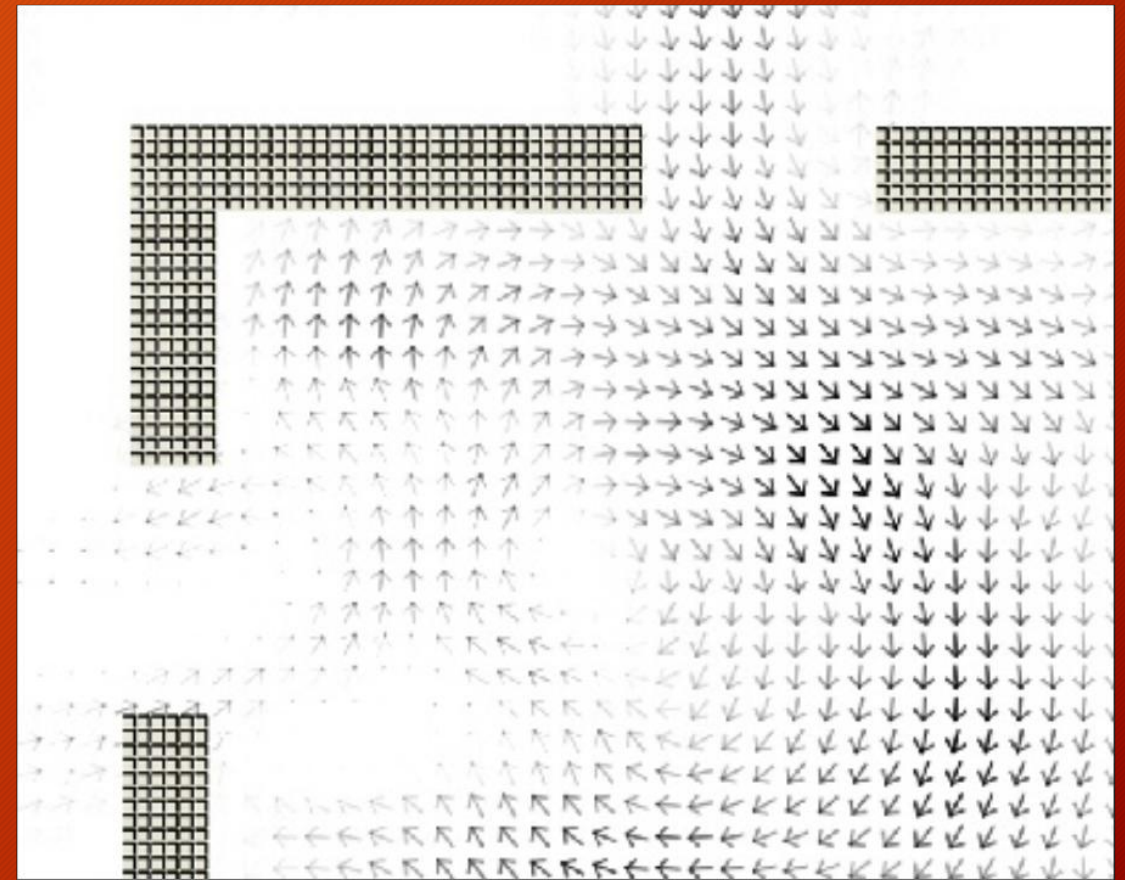
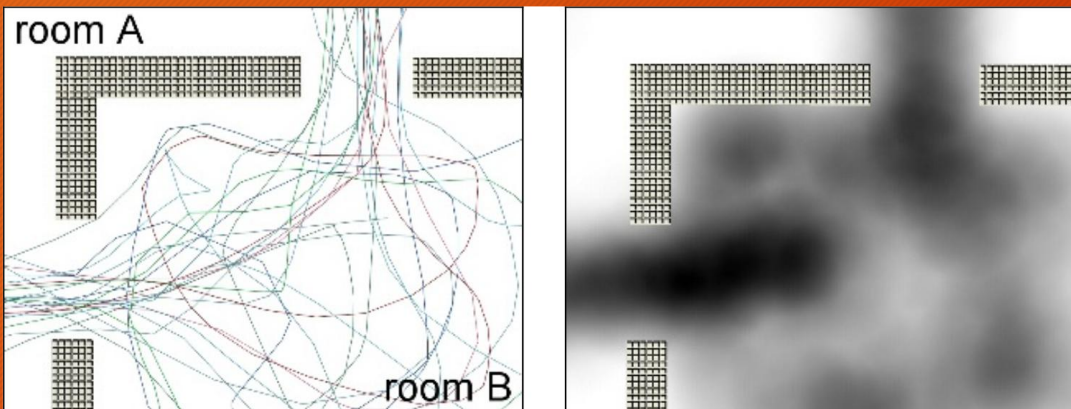
VU-Flow 3/6

- Time dependent visualization & seen areas



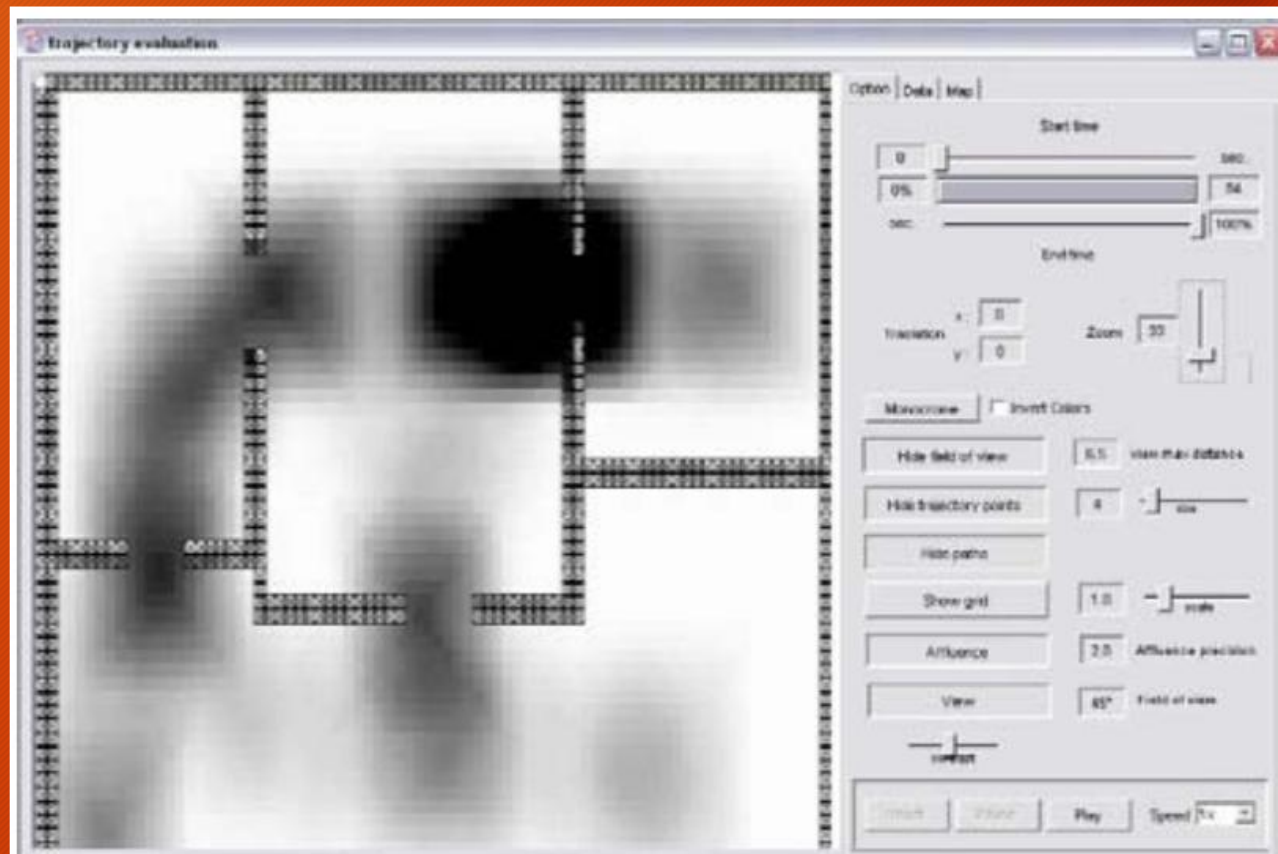
VU-Flow 4/6

- Users' predominant flow
 - Visualization displays information on predominant flow directions
 - For each cell of the map, the associated vector direction represents the predominant flow direction followed by users



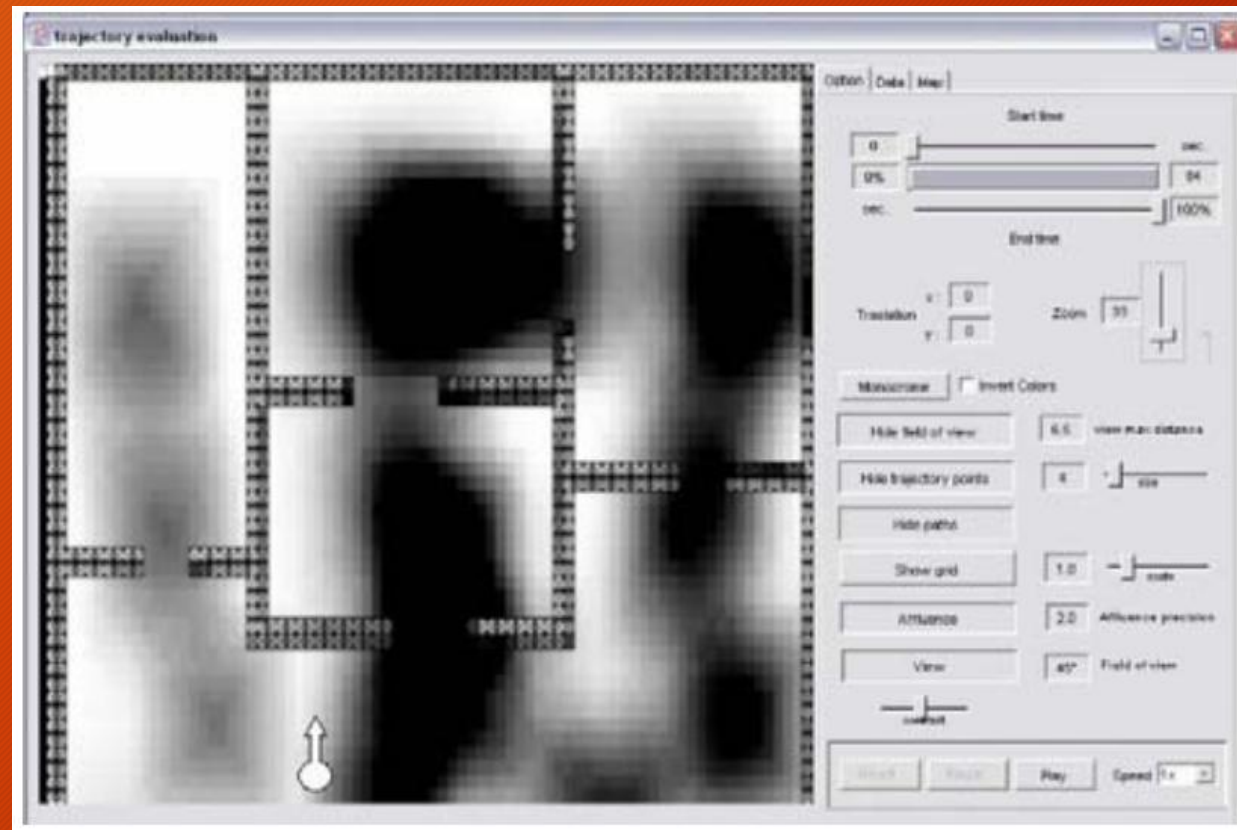
VU-Flow 5/6

- Identify navigation problems
 - Time-dependent visualization
 - Example: virtual exhibition with free navigation



VU-Flow 6/6

- Identify architecture influence on users' path
 - Time-independent visualization
 - Example: identify areas more/less visited in VE

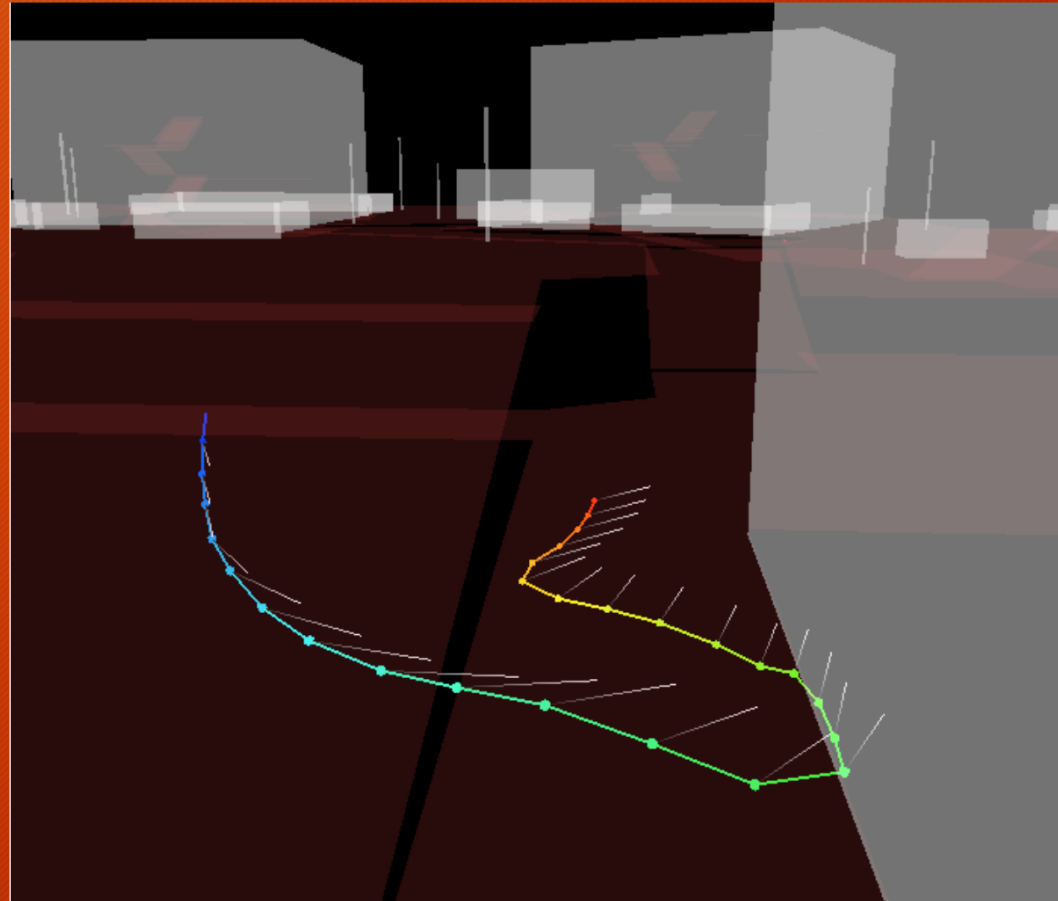


PlayerViz 1 / 7

- Tool proposed by Dixit & Youngblood, 2008
- Visual data mining and automated data processing
- Data considered: position & orientation over time
- Find occurrences of behavioral patterns:
 - Pirouettes
 - Flusters
 - Jumpers
 - Emergent behavior
 - Learning
- Context: Urban Combat Testbed project
- Dataset: 3079 players

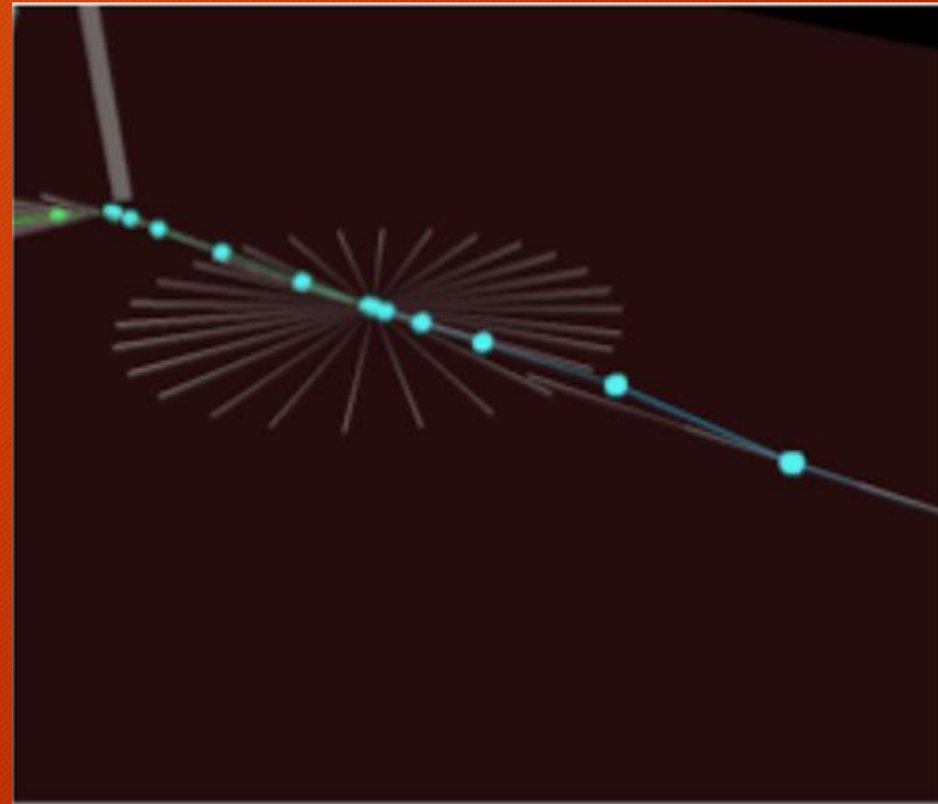
PlayerViz 2/7

- Player trace
 - Position: sphere
 - Orientation: oriented line segment
 - Path: spheres connected by line segments
 - Flow time: rainbow color scale blue-green-red normalized



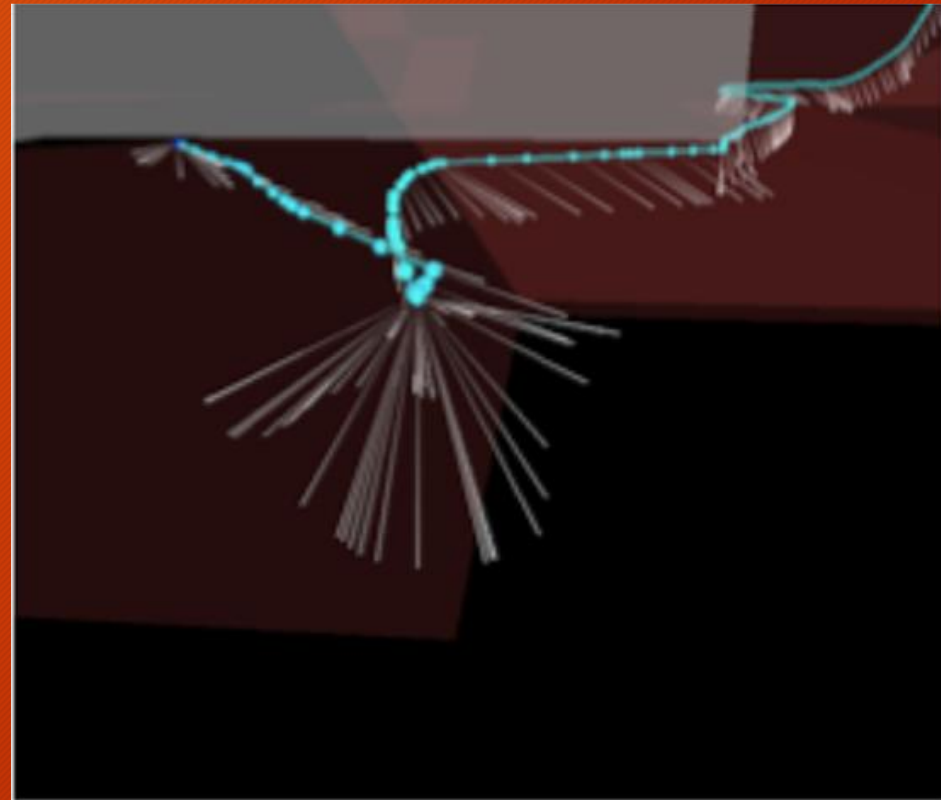
PlayerViz 3/7

- Deriving behavioral patterns: pirouette
- Behavior: the player turns nearly around 360° while standing still
- Causes:
 - The player is lost
 - The player is reorienting himself/herself
- When:
 - At the start of a play session
 - The player encounters an area with an ambiguous design
- Detection benefit: assist players by turning on a navigation arrow or highlighting the direction in which they need to go



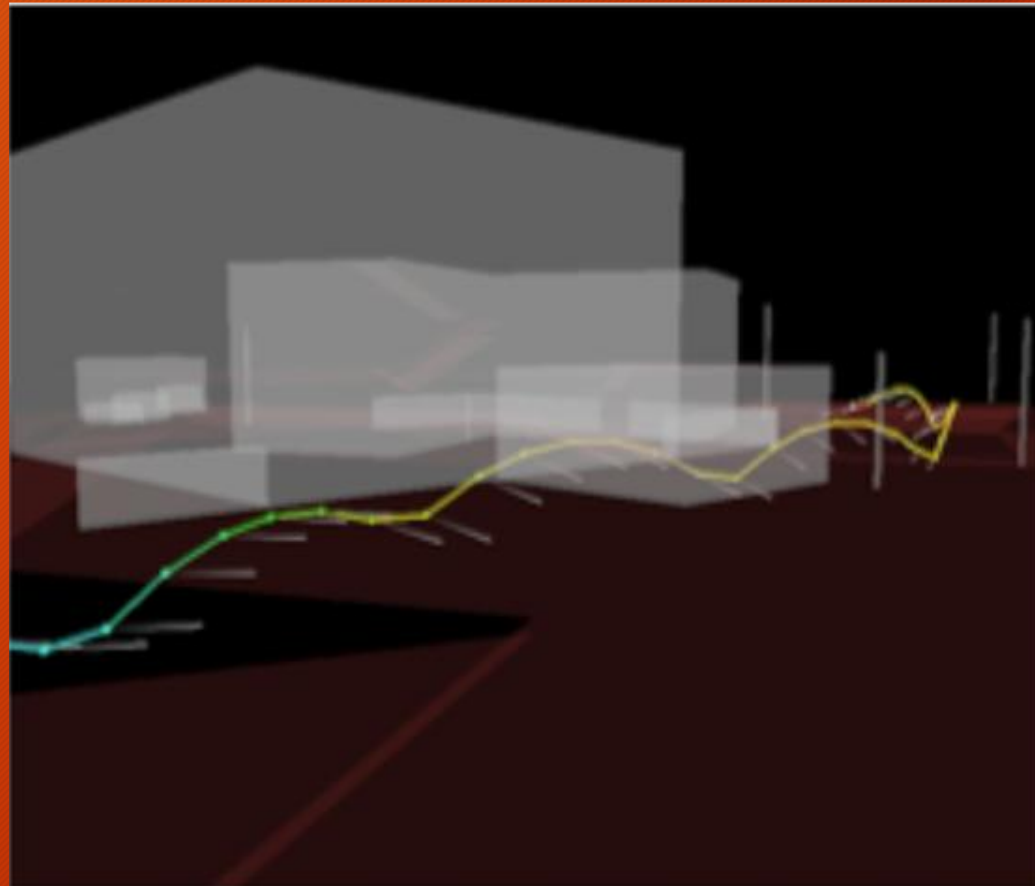
PlayerViz 4/7

- Deriving behavioral patterns: fluster
- Behavior: player's look direction gets stuck in a relatively confined cone with rapid changes in view for a period of time
- Causes:
 - Loss of mouse control / high mouse sensitivity
 - Gimbal lock in the player's rotation angles
- Detection benefit: Force players' look direction to a certain angle to help them escape the fluster



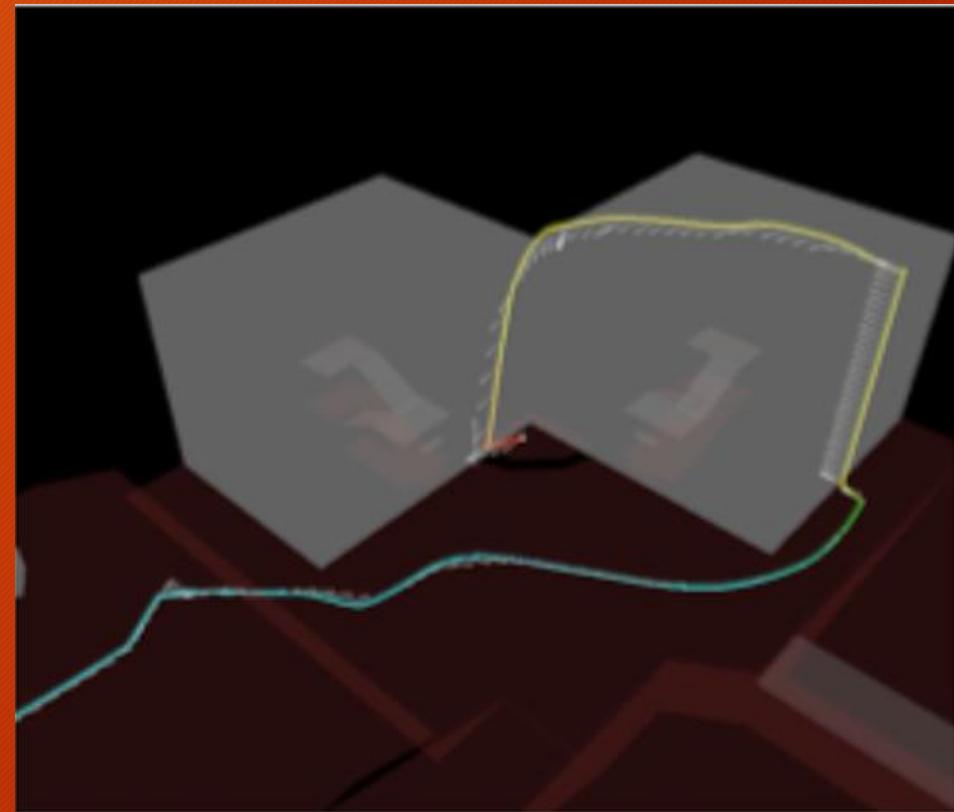
PlayerViz 5/7

- Deriving behavioral patterns: jumpers
- Behavior: the player constantly jumps through the VE
- Causes: advanced player's strategy to
 - Evade attack
 - Move faster through the VE
- Detection benefit: Compensate for increased speed



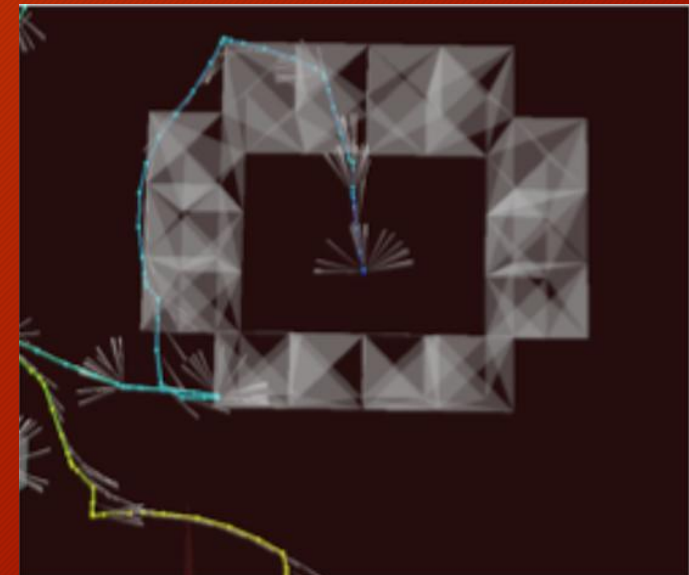
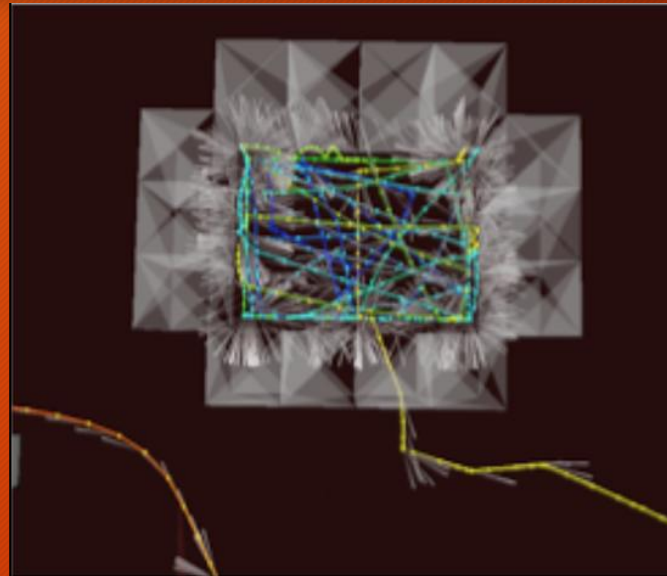
PlayerViz 6/7

- Behavior: players act in ways the developer did not anticipate
- Example: player climbed the side of the building, walked over the roof, then jumped down onto the goal
- Detection benefits:
 - Better understanding how players are interacting with the VE
 - Discover short-cuts that players might take



PlayerViz 7/7

- Deriving behavioral patterns: positive learning
- Behavior (e.g.):
 - The players' trace from their more recent play session is shorter than their previous one
 - The players did fewer pirouettes than in their previous play session
- Example:
 - First session: players realize much later that they can break crates to escape
 - Second session: players broke crates immediately and escaped
- Detection benefits: track learning across levels and given challenges



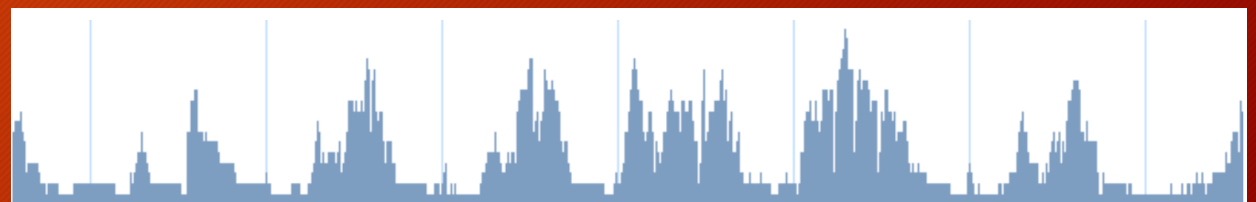
HeapCraft 1/5

- Proposed by Mueller et al., 2015
- Open-source suite of interactive data exploration and visualization tools
 - Epilog Dashboard
 - Map Miner
 - Classify
 - Graph Miner
- Context: Minecraft
- Data collected: player movement, block placement, mining, inventory content
- Aim: analyze and potentially influence player behavior in Minecraft
- Player behavior classification
- Quantifying and improving the collaboration of players

HeapCraft 2/5

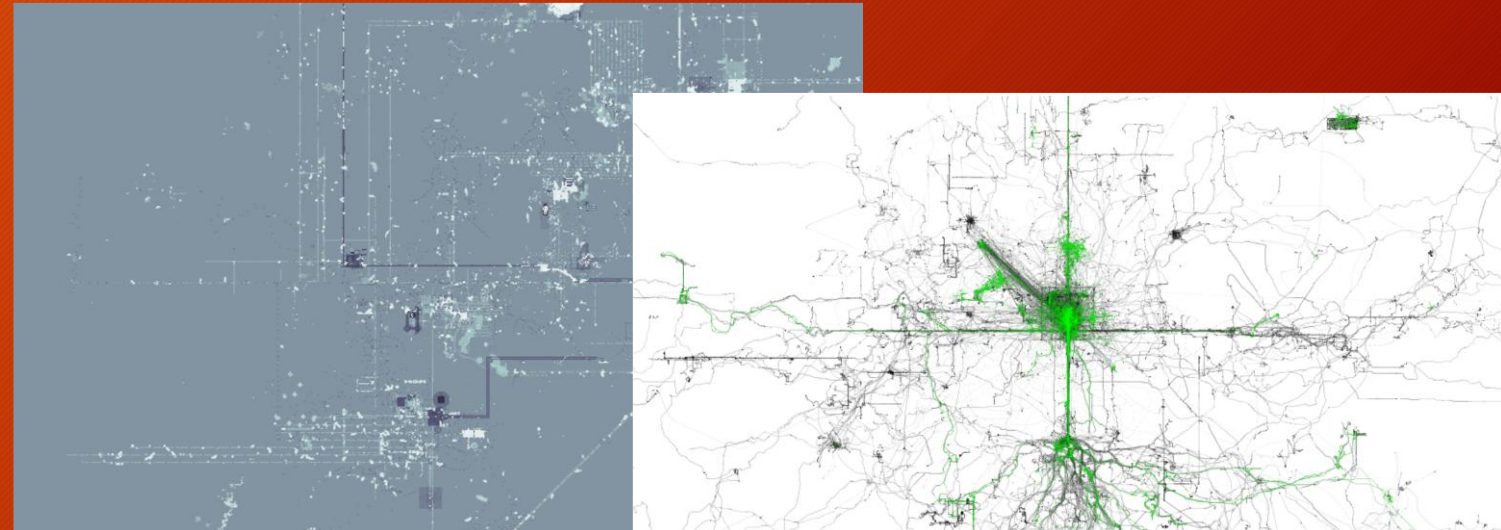
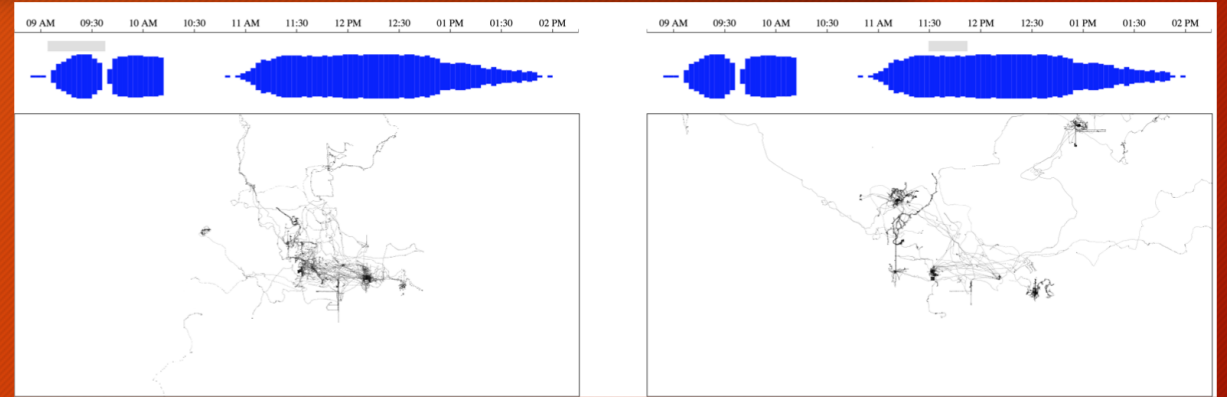
- Real-time summary statistics of behavioral data
- Tool: Epilog Dashboard
- Data for each player:
 - # placed blocks
 - total travel distance
 - total time active
 - last time active
 - collaboration index

Most Active Players						all players
Name	Last active	Active for	Traveled	Mined	Placed	Social
[redacted]	209d 22h	1d 16h	139,052	27,595	7,446	0.037
[redacted]	182d 9h	1d 12h	211,851	41,762	15,857	0.035
[redacted]	228d 4h	1d 0h	191,096	18,696	12,701	0.093
[redacted]	84d 20h	23h 41m	186,811	15,448	7,349	0.304
[redacted]	134d 23h	22h 3m	217,126	11,140	5,558	0.193
[redacted]	232d 2h	16h 27m	93,127	13,926	9,882	0.057
[redacted]	164d 22h	16h 25m	123,052	10,267	6,246	0.333
[redacted]	228d 3h	16h 4m	109,793	8,816	5,794	0.138
[redacted]	231d 1h	16h 3m	100,770	11,200	7,117	0.170
[redacted]	161d 23h	15h 46m	101,716	9,369	5,222	0.401



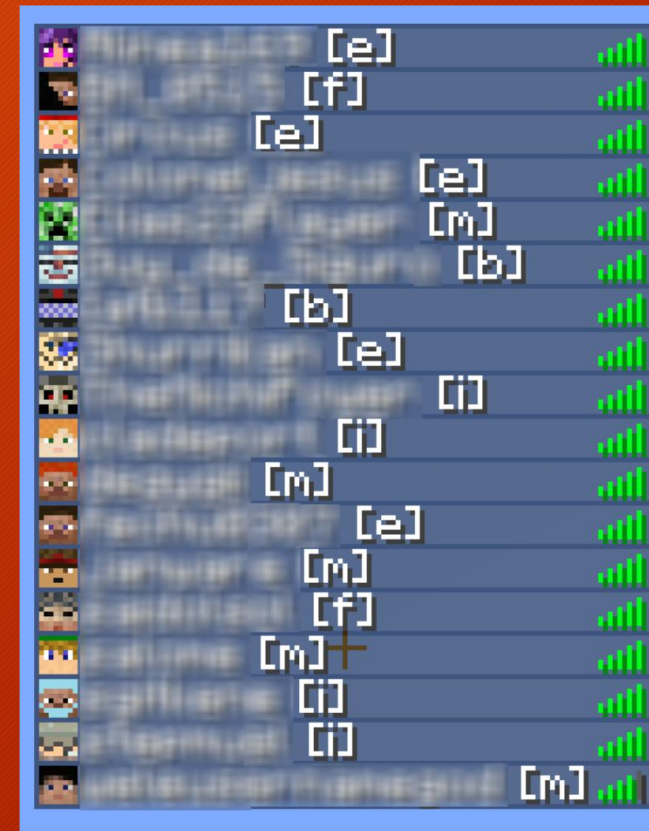
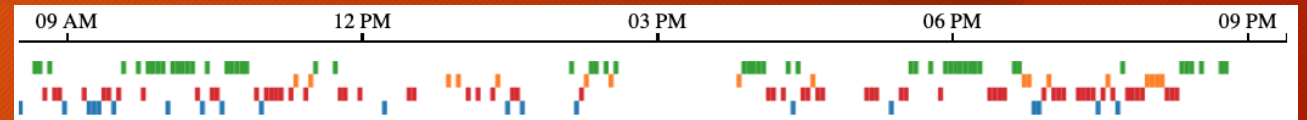
HeapCraft 3/5

- Explore data spatially and temporally
- Tool: Map Miner
- Visualizes timelines with heatmaps
- Spatial data for each player:
 - Positions in the map
 - Blocks removed/added
 - Player contact



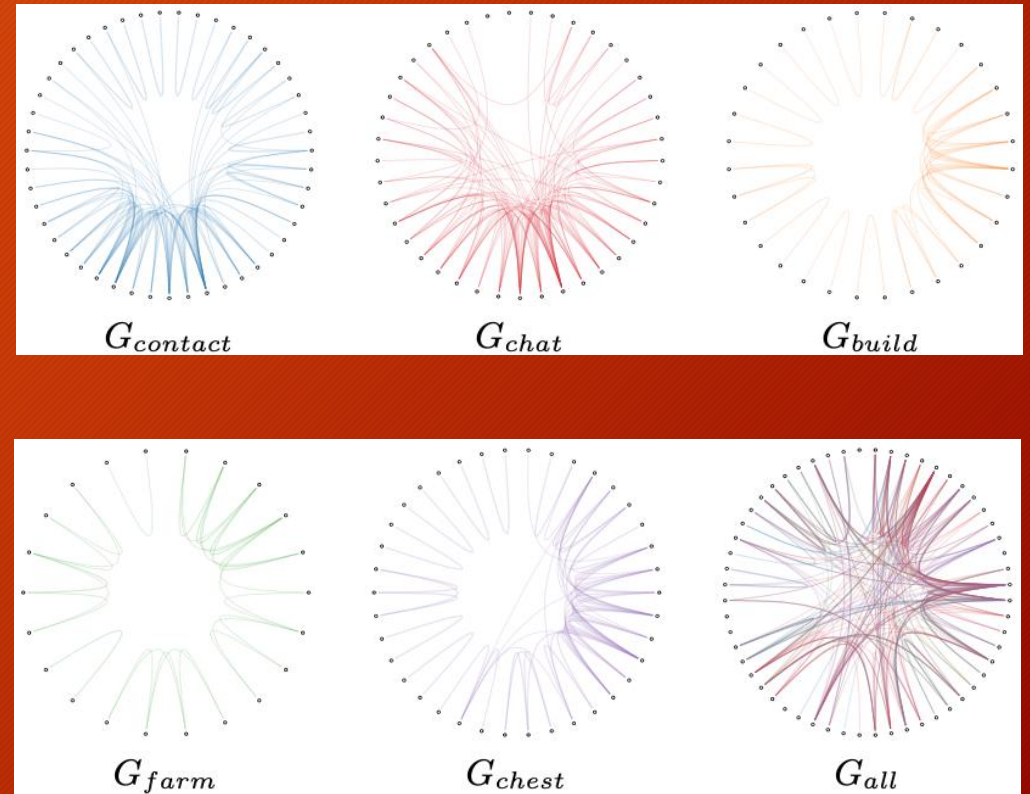
HeapCraft 4/5

- Analyze behavior of all online players
- Tool: Classify
- Each player is classified based on his current behavior into categories:
 - Build
 - Mine
 - Fight
 - Explore
 - Idle
- Aim: allow players to identify similar players for potential collaboration



HeapCraft 5/5

- Inspection of player interaction and collaboration
- Tool: Graph Miner
- Indicates if players share activities
- Collaboration indicators:
 - Player contact
 - Chat between players
 - Shared buildings
 - Shared farms
 - Shared chest

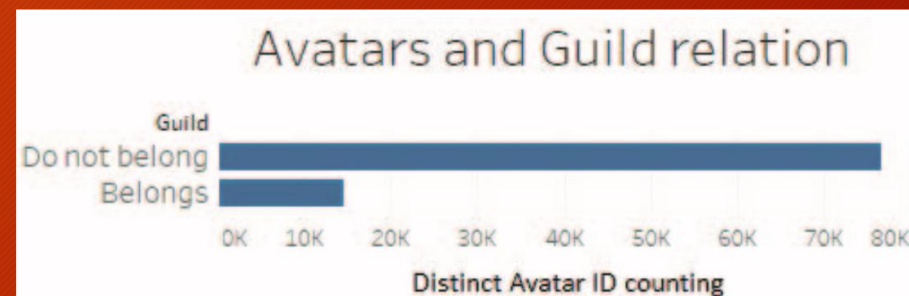
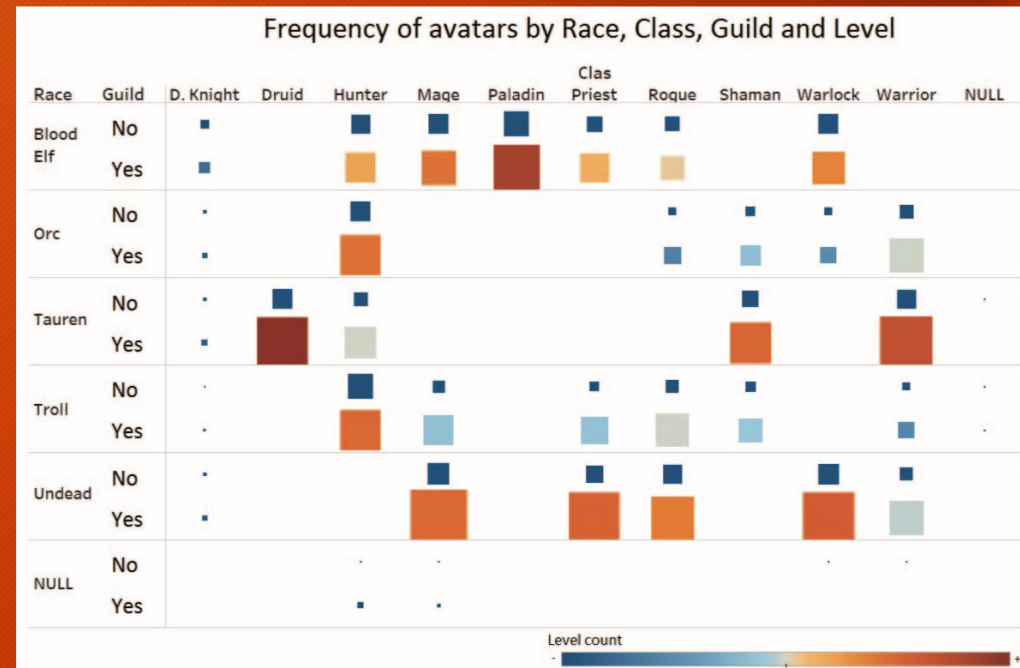


CDH for pattern recognition 1 / 6

- Proposed by Barros and Notargiacomo, 2016
- Platform: Cloudera Distribution for Hadoop
- Tool for data visualization: Tableau
- Database: World of Warcraft Avatar History (WoWAH)
- Aim:
 - 1. Players' characteristics:
 - Count frequency of appearances by Race, Class, Level & Guild membership
 - Heatmap of most/less frequented zones
 - 2. Avatar's gameplay time

CDH for pattern recognition 2/6

- Frequency of appearances & guild membership
- Relation between attractiveness and guild participation
- Most assiduous combinations:
 - Undead Mage
 - Tauren Druid
- Relation number of avatars that belong vs do not belong to guilds



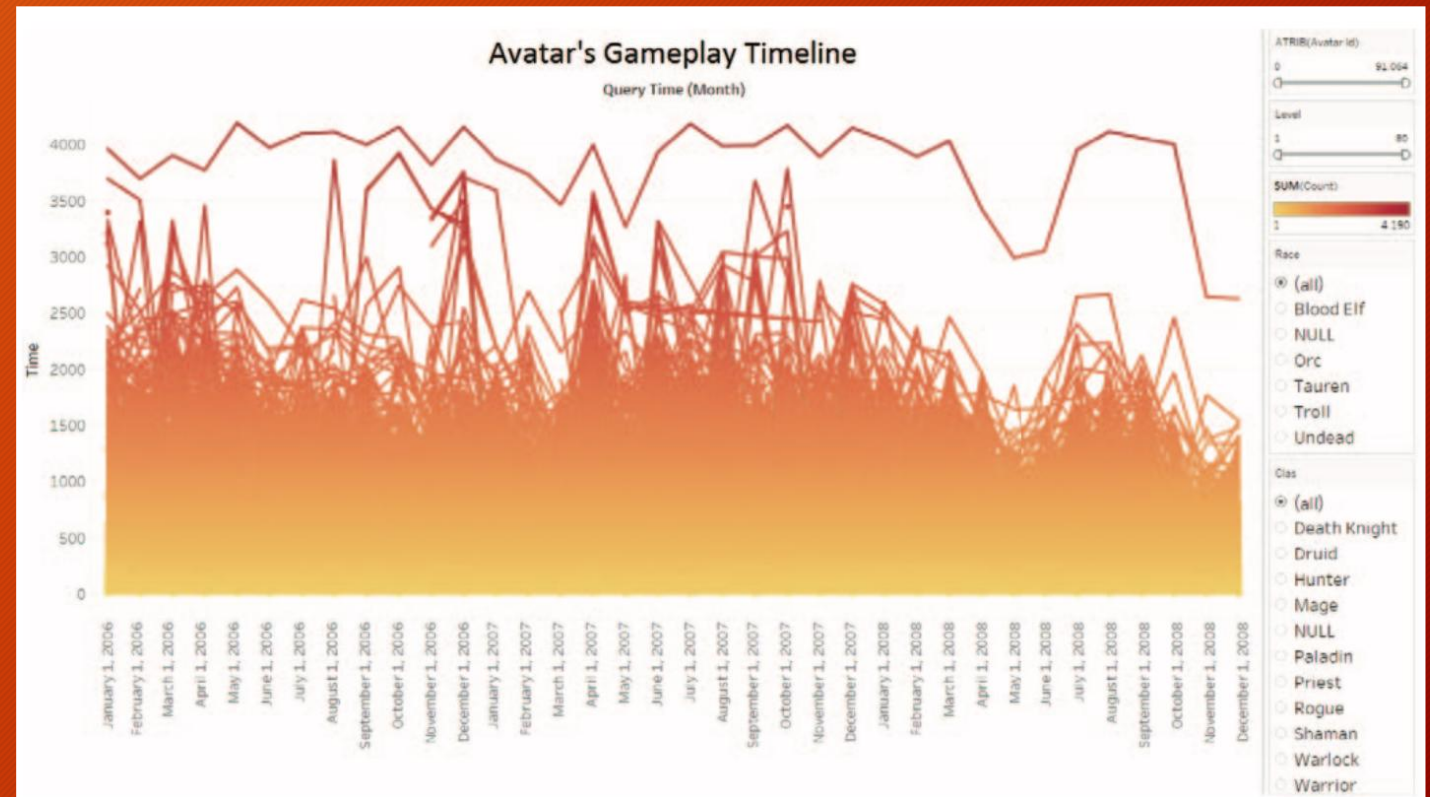
CDH for pattern recognition 3/6

- Heatmap of frequented Zones
- Most/less frequented Zones in real-time
 - Zones less frequented: dark blue
 - Zones most frequented: orange to red
- Filtered by Race, Class, Level



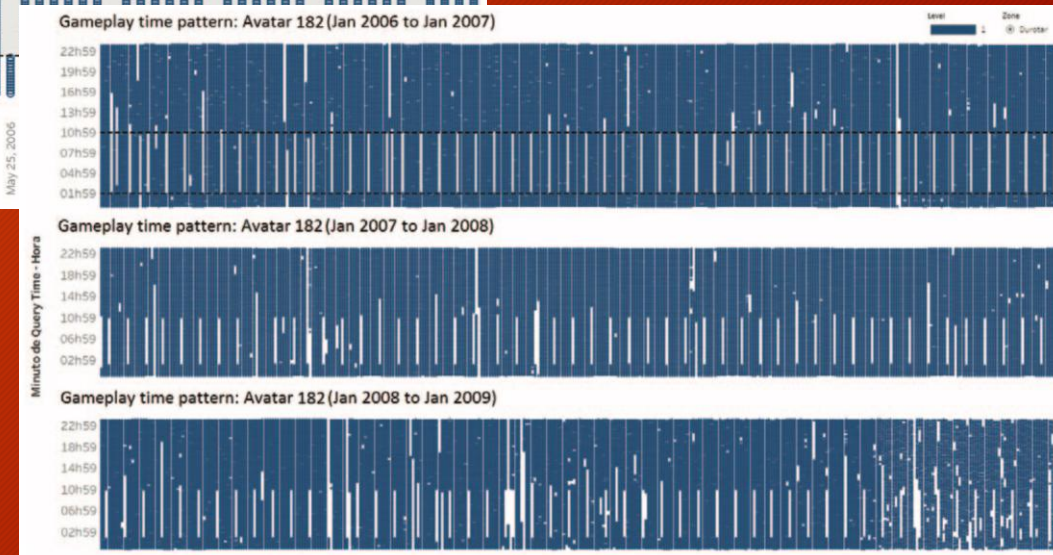
CDH for pattern recognition 4/6

- Avatar's gameplay time
- # hours spent by all players individually
- Identified outlier:
 - played 635 hours/month
 - online 88% of the time



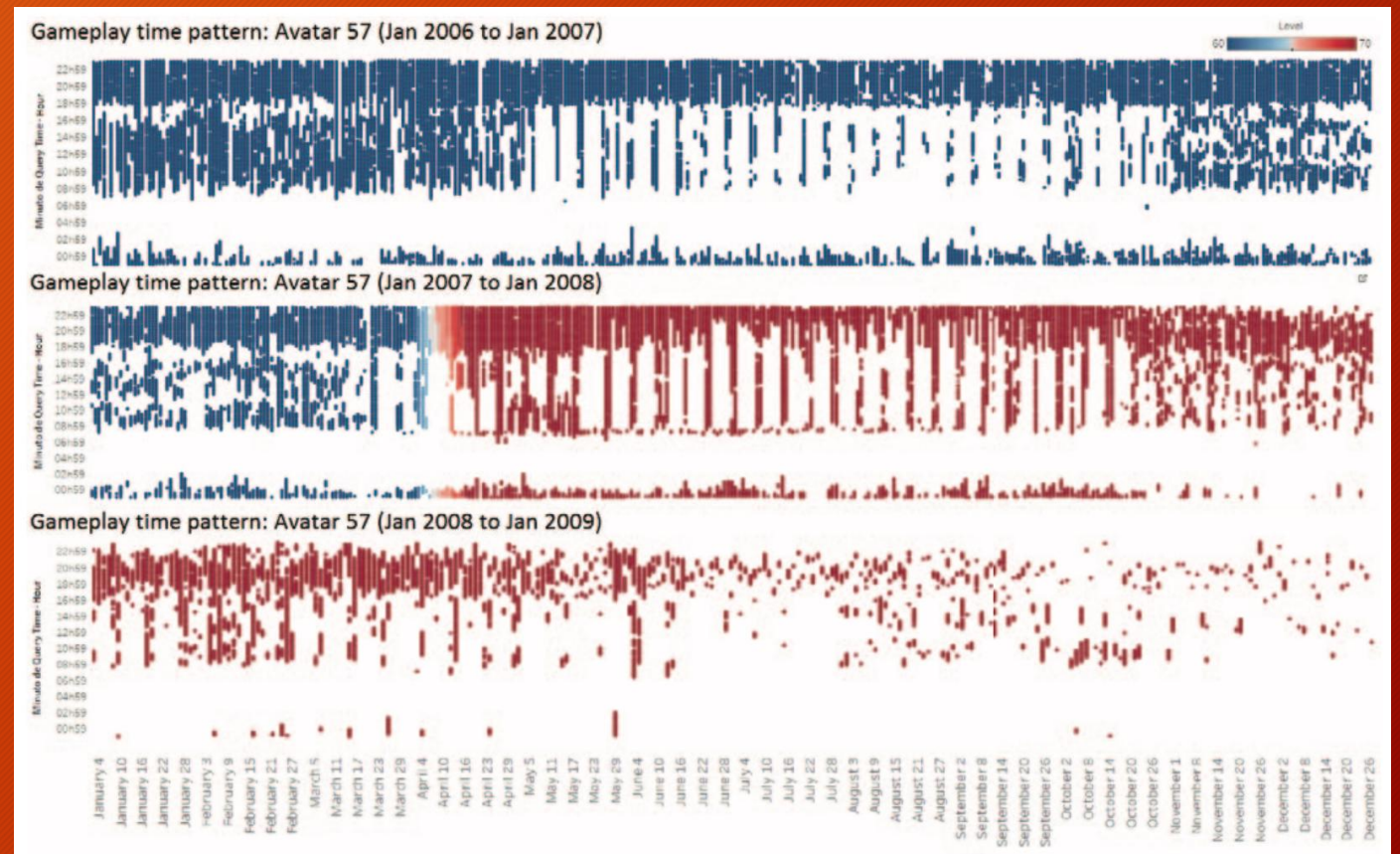
CDH for pattern recognition 5/6

- Analysis of the first most assiduous avatar
- Avatar 182 gameplay time pattern
- Always in level 1
- Not belong to any guild
- Not changed any zone
- Detected login/logout patterns



CDH for pattern recognition 6/6

- Analysis of the second most assiduous avatar
- Avatar 57 gameplay time pattern
- Gameplay time not as linear as avatar 182
- Assiduity more intense by end of 2006 and 2007 than end of 2008
- Color transition: time of expansion release
- Level up from 60 to 70

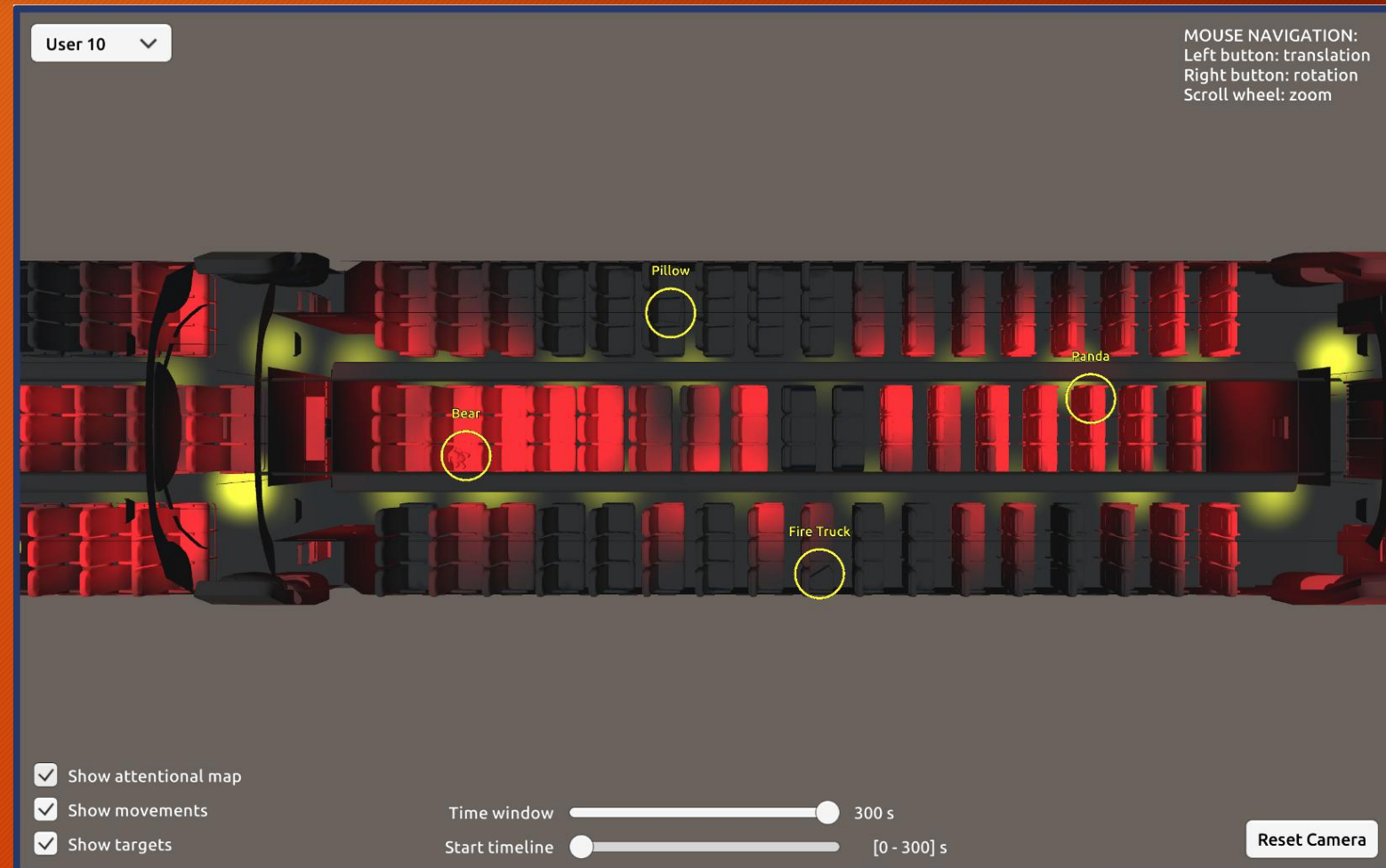


User's behavior in indoor 3D VEs 1/13

- Tool proposed by Chittaro & Serafini, 2021
- Aims at helping the analyst to understand user behavior
- Visualizes usage data directly within the VE, viewable from any angle
- Combines a 3D heatmap of looked-at areas with a heatmap of movement

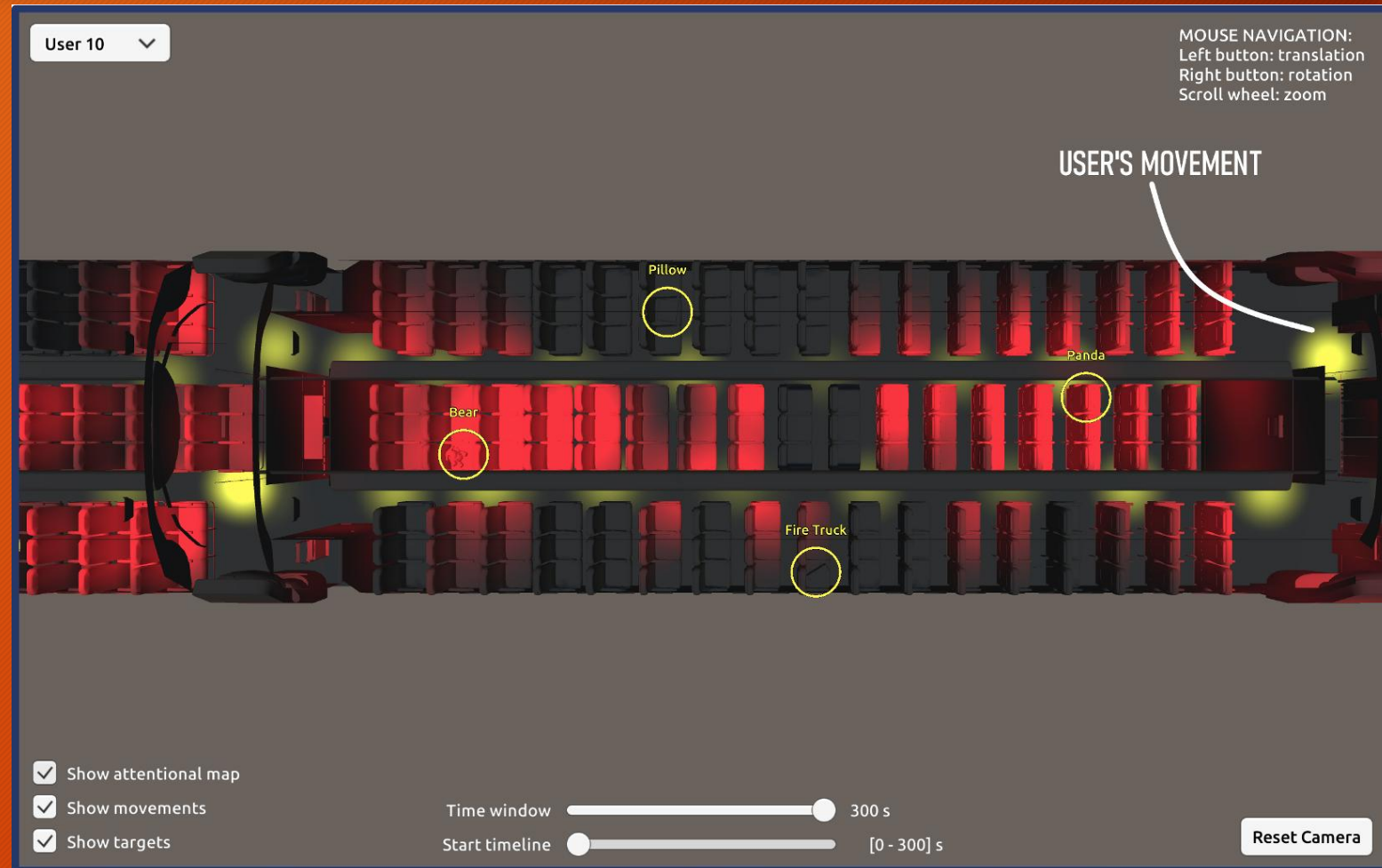
User's behavior in indoor 3D VEs 2/13

- Example



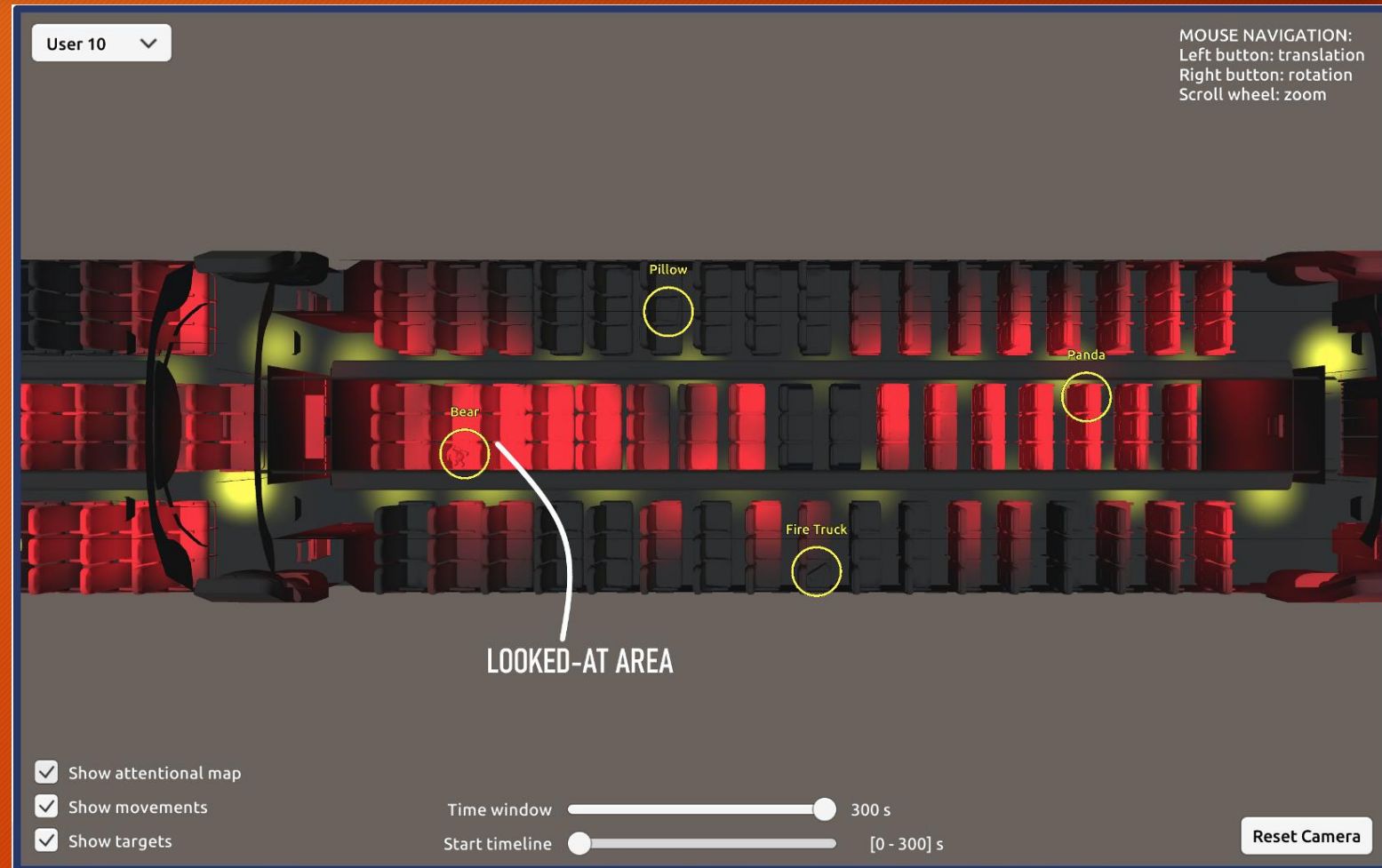
User's behavior in indoor 3D VEs 3/13

- Example



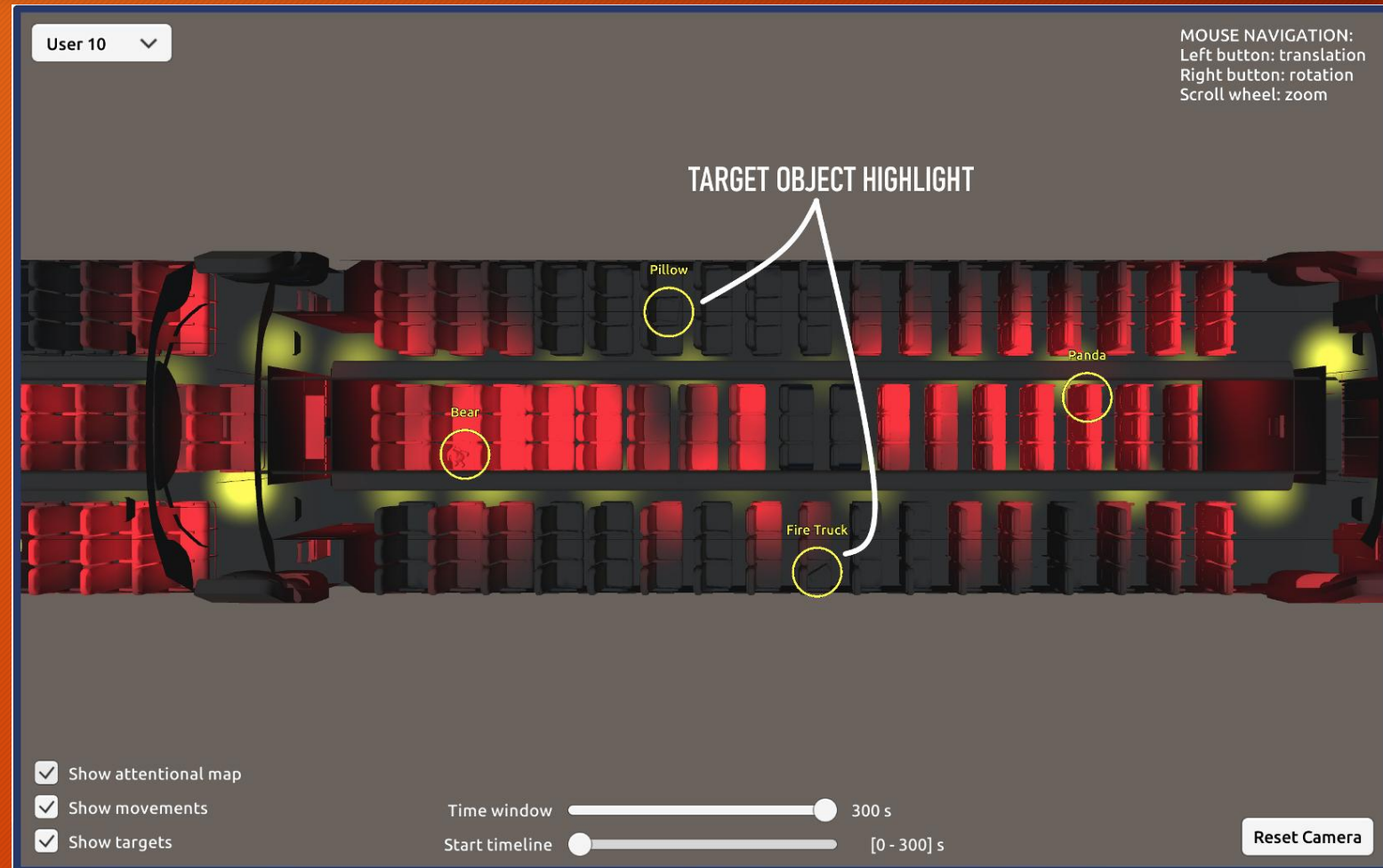
User's behavior in indoor 3D VEs 4/13

- Example



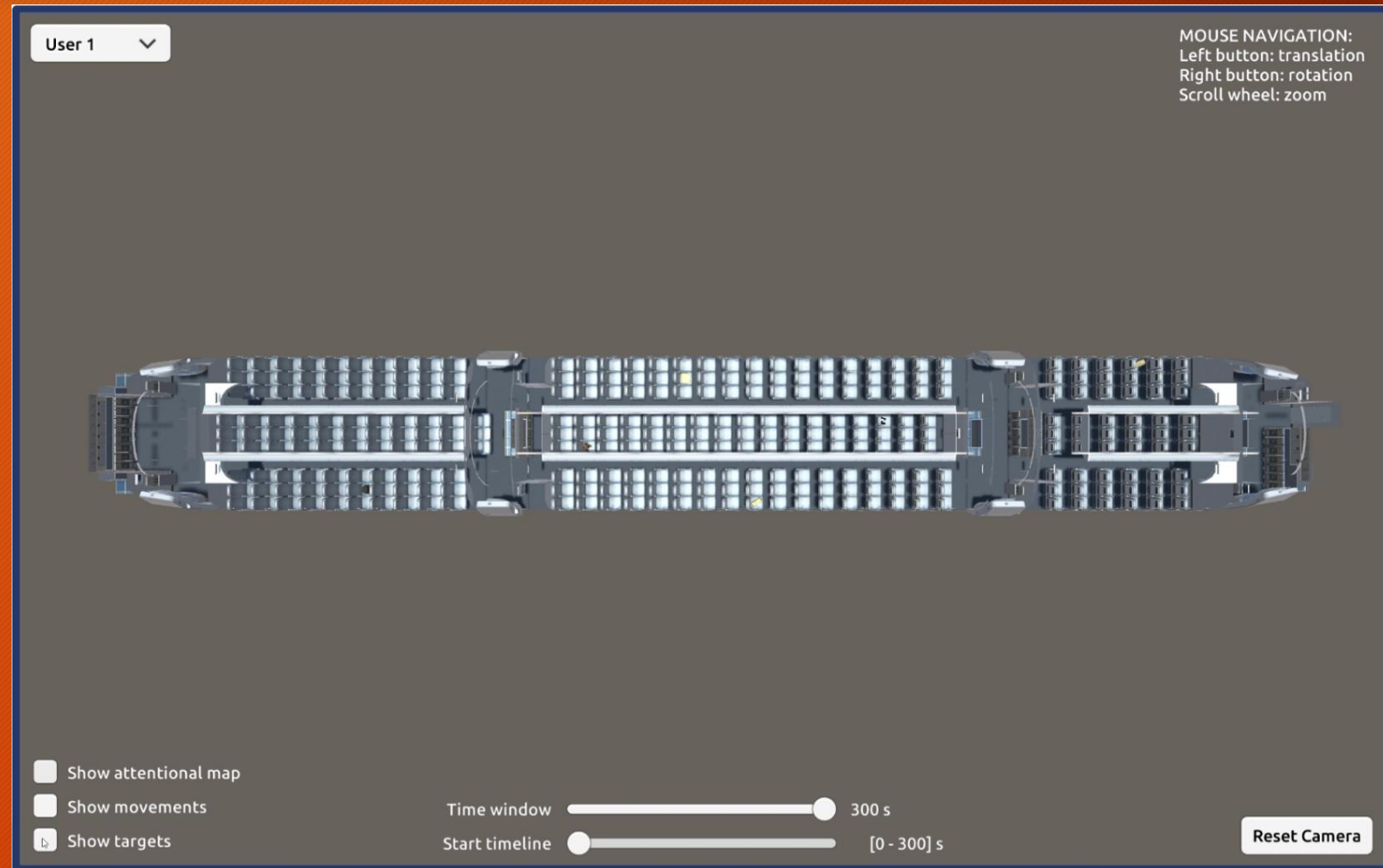
User's behavior in indoor 3D VEs 5/13

- Example



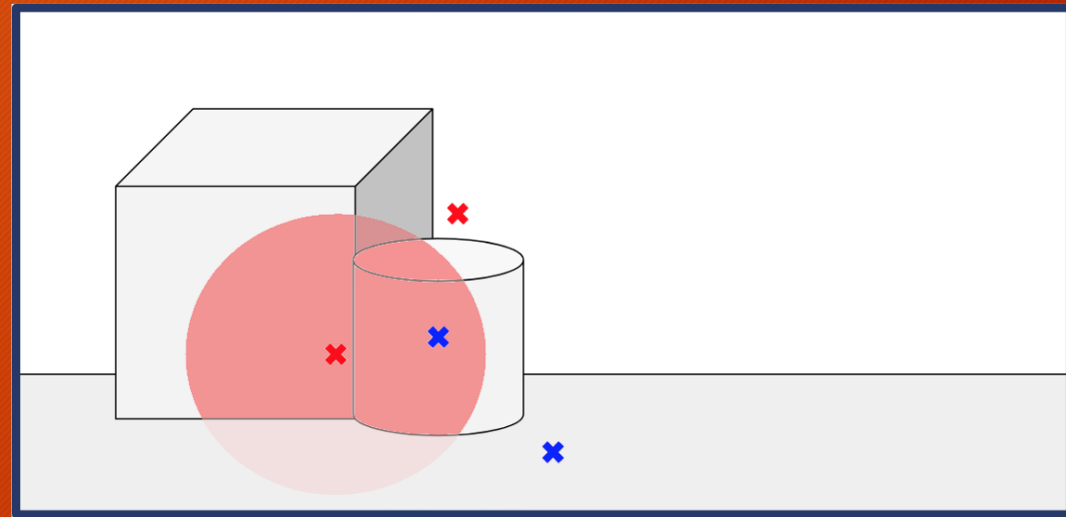
User's behavior in indoor 3D VEs 6/13

- Video example



User's behavior in indoor 3D VEs 7/13

- 3D heatmap
- Looked-at objects are identified using a raycast pointing to the center of the scene viewed by the user
- A circular area is colored on the surface of the object around the looked-at point and objects in its close proximity

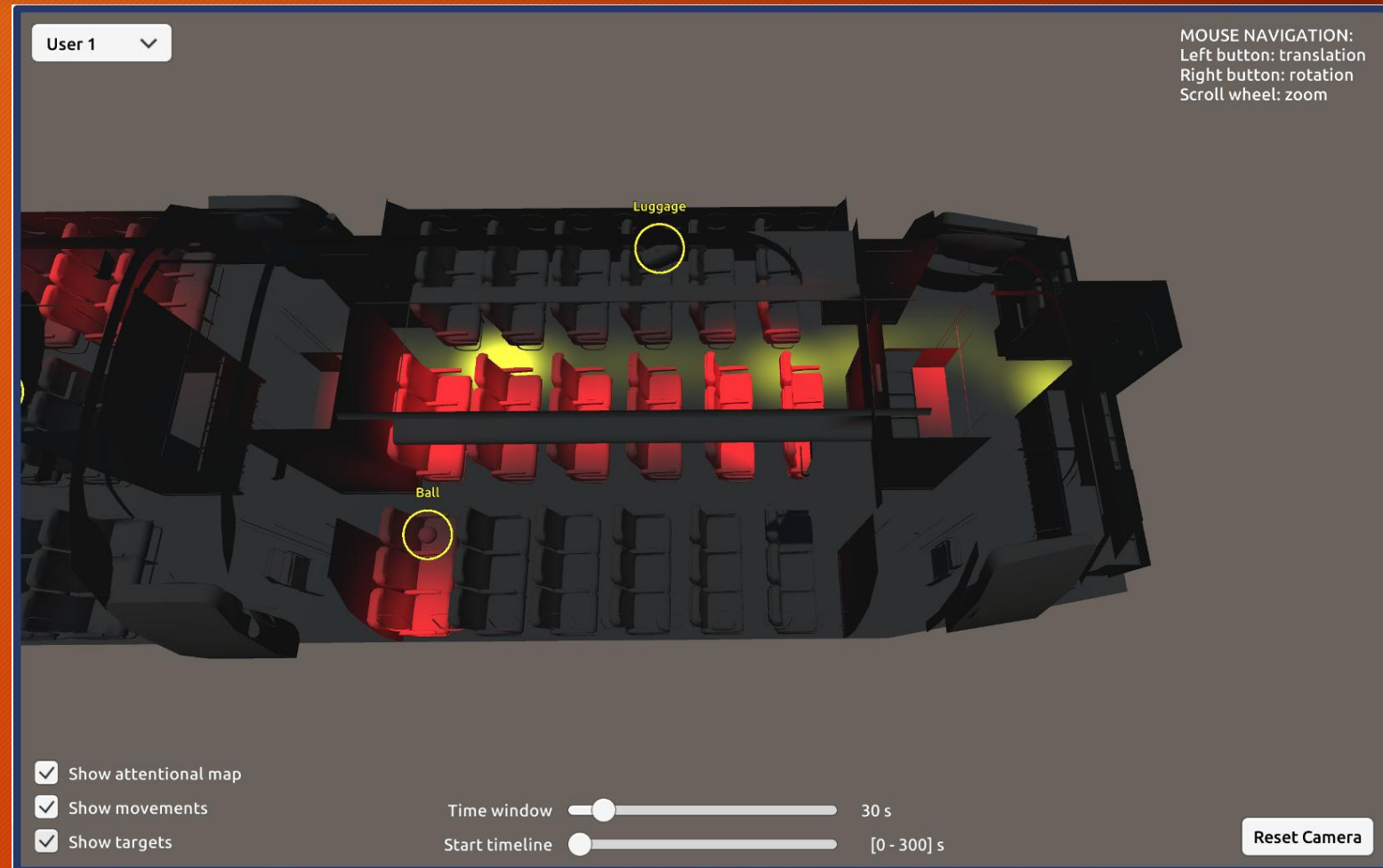


User's behavior in indoor 3D VEs 8/13

- The tool does not need an eye tracker
- The analyst can:
 - Choose the objects of interest
 - Turn on or off each visualization
 - Apply a temporal filter to visualizations

User's behavior in indoor 3D VEs 9/13

- Example



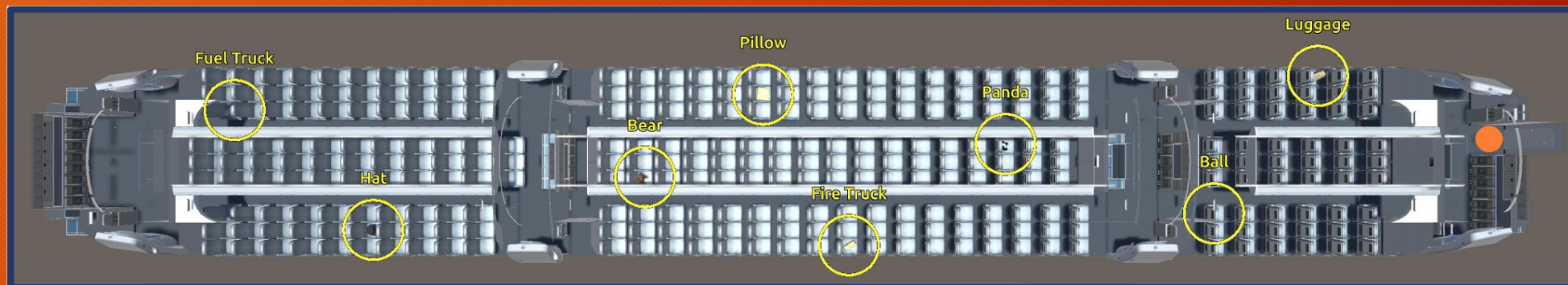
User's behavior in indoor 3D VEs 10/13

- Pilot study
- Dataset: behavioral data of 127 real users from an experiment that tested their ability to remember the position of target objects in a VE
- VE: Boeing 777 twin-aisle aircraft
- Participants: four experts in the field of VEs
- Tasks: four analysis tasks concerning the behavior of 10 representative users taken from the dataset

User's behavior in indoor 3D VEs 11/13

- Tasks

- 1. Recognize the user's navigation pattern
- 2. Detect whether the user followed a specific path
- 3. Detect whether the user had explored a specific wide area of the VE
- 4. Identify which objects were looked at and which were not by the user



User's behavior in indoor 3D VEs 12/13

- T1 & T2: all experts answered correctly for all users
- T3: experts answered correctly for a number of users that ranged from 9 to 10 out of 10
- T4: experts identified correctly a number of objects that ranged from 72 to 80 out of 80

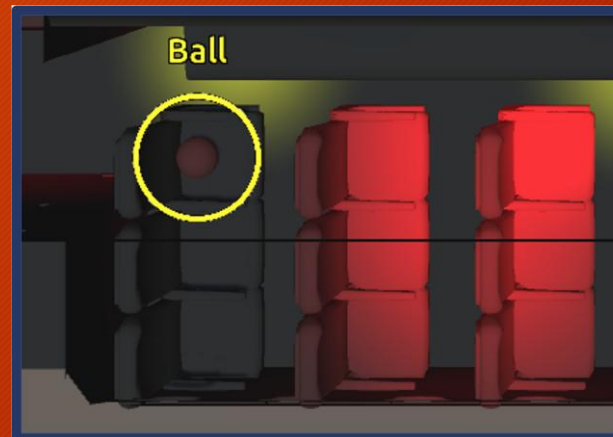
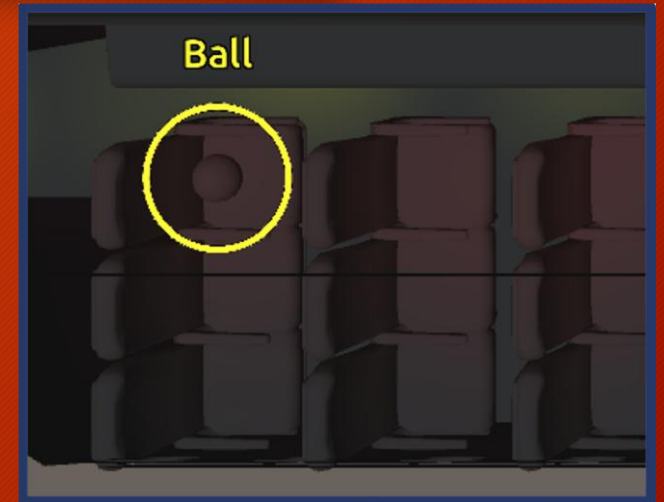
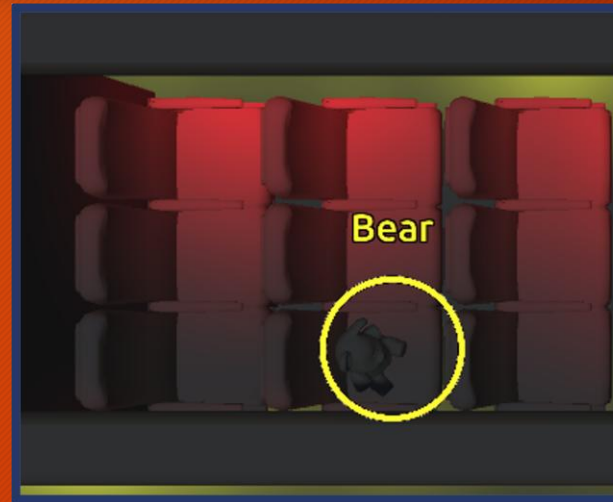
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	T3	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10	
E1												$p < 0.01$
E2												$p < 0.01$
E3												$p < 0.05$
E4												$p < 0.05$

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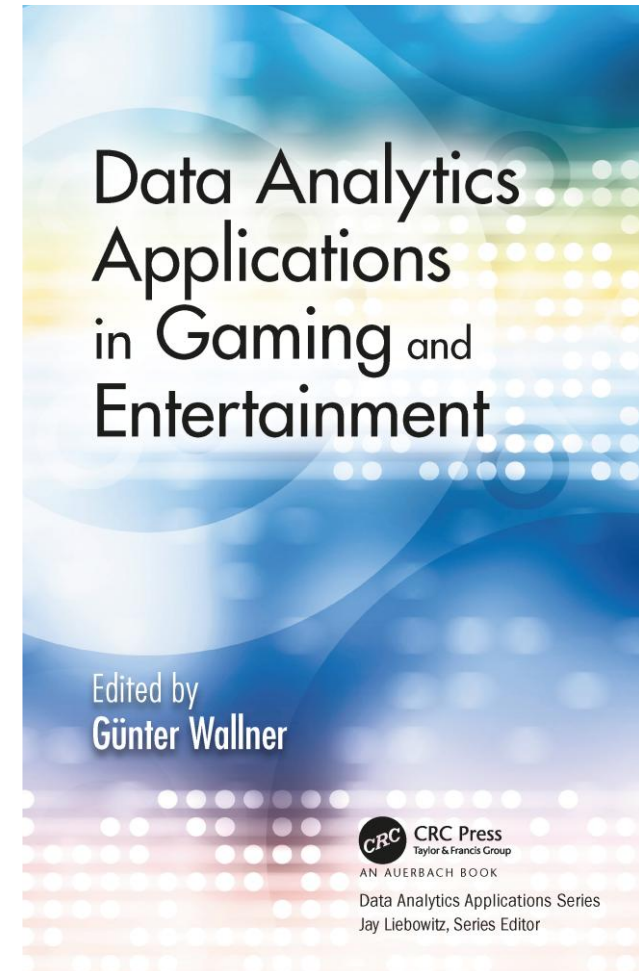
User's behavior in indoor 3D VEs 13/13

- Types of errors in T4
- Erroneously identifying a target object as looked at
- Erroneously identifying a target object as not looked at
- Erroneously identifying a small object as not looked at when it is placed on a large object not looked at



Bibliography 1 / 3

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